PG & INI-CET SMART FLASHCARD APP

Product Requirements Document (PRD) – For Development Team

Purpose: A medical exam-preparation flashcard app focused on active recall, image recognition, MCQ practice and rapid revision for PG/INI-CET preparation.

1. PRODUCT VISION

Build a fast, simple and exam-oriented flashcard application for medical students. The app should make it easy to convert study material into active-recall cards and revise those cards repeatedly.

- Target users: MBBS students preparing for PG entrance examinations and INI-CET.
- Primary use cases: image identification and question-based MCQ revision.
- Cards must remain permanently available for later revision.
- Creation and revision are separate processes: create once, revise many times.

2. CORE FEATURES – MUST HAVE

A. IMAGE FLASHCARD MODE

- User uploads or captures an image.
- The image becomes the front of the flashcard.
- User enters the correct answer/diagnosis/label on the back.
- During revision, the image is shown first and the answer stays hidden.
- User taps 'Reveal Answer' to see the stored answer.
- Optional back-side fields: explanation, key findings, differential diagnosis and notes.
- Image must remain clear enough for clinically relevant details.

Example: Upload X-ray → Front: X-ray image → Back: diagnosis + key finding.

B. QUESTION → AUTOMATIC MCQ MODE

- User enters/pastes a question.
- User provides ONE correct answer.
- AI generates THREE plausible but incorrect distractors.
- Final MCQ contains exactly 4 options: 1 correct + 3 distractors.
- Correct answer is stored separately and must not be identifiable before submission.
- During revision, user selects one option and taps Submit/Check Answer.
- After submission, show Correct/Incorrect and identify the correct answer.
- Optional: show a short explanation after submission.
- User can edit AI-generated distractors before saving.

AI rule: The user-provided answer is the source of truth. AI generates distractors around it and must not silently replace it.

C. SAVED FLASHCARD LIBRARY

- Every completed card is automatically saved.
- Cards can be organized by subject, topic, exam, custom deck and tags.
- Suggested subjects: Medicine, Surgery, OBGYN, Pediatrics, Anatomy, Physiology, Pathology, Pharmacology, Microbiology, Ophthalmology, ENT, Orthopedics, Dermatology, Radiology and PSM.
- Search by question, answer, tag or subject.
- Filter by Image/MCQ, subject, topic, difficulty and revision status.

- Allow edit, duplicate, archive and delete.

D. REVISION WITHOUT RE-ENTERING THE QUESTION

- Critical requirement: a saved flashcard must be revisable anytime without entering the original question again.
- Image card: show image → user recalls → Reveal Answer.
- MCQ card: show question + 4 options → user answers → Submit → result/explanation.
- User can revise cards individually, by deck, subject or revision queue.

3. USER FLOW

HOME SCREEN

- Create Flashcard
- My Flashcards
- Revision / Due Cards
- Subjects / Decks
- Search
- Progress / Statistics

CREATE FLASHCARD

- Choose Image Card OR Question Card.
- Image: Upload/Capture → Enter Answer → Optional Explanation/Tags → Save.
- Question: Enter Question → Enter Correct Answer → AI generates 3 distractors → User reviews/edits → Save.
- Card is added to the selected deck and becomes available for revision.

REVISION FLOW

- Open Revision and select a deck/subject or 'All Due Cards'.
- Image card: display image only → user thinks → Reveal Answer.
- MCQ card: display question + 4 options → select → Submit.
- Show result → optional explanation → mark Known / Needs Revision.
- Continue to next card.

4. FLASHCARD DATA STRUCTURE

Field	Purpose
Card ID	Unique identifier
Card Type	IMAGE or MCQ
Question/Prompt	MCQ question; optional prompt for image card
Image	Uploaded image for image-based card
Correct Answer	User-provided answer; source of truth
Option A-D	Four MCQ options; one correct, three distractors
Explanation	Optional answer explanation/key points
Subject / Topic	Classification for filtering
Tags	Custom searchable labels
Difficulty	Easy / Medium / Hard
Created At	Creation timestamp
Last Reviewed	Latest revision timestamp
Review Status	New / Learning / Known / Needs Revision
Review Count	Number of revisions

5. AI MCQ GENERATION REQUIREMENTS

Input: Question + Correct Answer → Output: Three plausible distractors + optional explanation.

- Distractors should be from the same conceptual category as the correct answer.
- Avoid giveaway patterns such as one option being much longer or more detailed.
- Avoid duplicate or synonymous options.

- Never change the user's supplied correct answer.
- If AI is uncertain, flag the card for user review instead of confidently inventing facts.
- User must be able to edit every AI-generated option.

Recommended pipeline: Question + Correct Answer → AI generation → uniqueness/plausibility validation → user preview → Save.

6. REVISION & SPACED-REVIEW SYSTEM

- MVP: manual ratings such as Know / Need Revision / Again.
- Preferred: automatic spaced-repetition scheduling.
- Suggested buttons: Again / Hard / Good / Easy.
- Again = show sooner; Easy = show later.
- Show 'Due Today' count on Home.
- Provide 'Review Missed Cards'.
- Maintain review history without deleting the card.

7. UX REQUIREMENTS

- One-hand friendly and minimal-tap interface.
- Large, clear image display.
- Answer stays hidden until explicit reveal/submission.
- Fast next-card navigation.
- Optional swipe navigation.
- Dark mode recommended.
- Auto-save drafts if user leaves creation midway.

8. IMAGE HANDLING

- Support JPG, JPEG, PNG and common mobile image formats.
- Allow camera capture and gallery upload.
- Allow crop and rotate.
- Compress intelligently while preserving diagnostic detail.
- Securely store image and permanently associate it with its card.

9. SEARCH, FILTER & DECKS

- Global search across questions, answers, explanations and tags.
- Custom decks such as 'INI-CET Rapid Revision', 'Radiology Images', 'OBGYN', 'Medicine ECGs'.
- Move/copy cards between decks.
- Filter by type, subject, topic, difficulty and review status.
- Sort by recently created, recently reviewed, due and most-missed.

10. PROGRESS & ANALYTICS

- Total cards created.
- Cards revised today and due cards.
- MCQ correct percentage.
- Most-missed topics.
- Image recall accuracy.
- Revision streak.
- Subject-wise progress.
- Optional weekly/monthly dashboard.

11. BACKEND / TECHNICAL REQUIREMENTS

- Persistent database so cards survive app restarts and updates.
- User authentication and cloud sync recommended.
- Images stored securely, separate from card metadata where appropriate.
- AI API keys must remain on a secure backend, not inside the mobile app.
- Retry/error handling for AI failures.
- Manual option entry must remain available if AI generation fails.

12. SAFETY & CONTENT CONTROL

- Educational tool only; not a substitute for clinical judgment.
- AI-generated explanations/distractors must be editable.
- Never silently overwrite user-entered correct answers.
- Allow users to report incorrect AI content.
- Protect uploaded study material and personal data.
- Backup/export functionality is recommended.

13. MVP – FIRST VERSION

- Login/profile.
- Create Image Flashcard.
- Create Question Flashcard.
- AI generates 3 distractors from one supplied correct answer.
- Permanent card storage.

- My Flashcards library.
- Image recall + reveal.
- MCQ solving + instant result.
- Edit/delete.
- Basic subjects/decks.
- Basic revision history.
- Search and filters.

14. PHASE 2

- Spaced repetition algorithm.
- Automatic daily revision queue.
- OCR from screenshots/PDF pages.
- AI-generated explanations.
- Automatic tagging and subject classification.
- Bulk import of questions/images.
- Analytics dashboard.
- Cloud sync across devices.
- Backup/export.
- Timed exam-mode MCQ sessions.

15. EXAMPLE USE CASES

Radiology: Upload X-ray → diagnosis on back → later show only X-ray → recall → reveal.

Dermatology: Upload lesion image → diagnosis on back → image-first revision.

Medicine MCQ: Enter question + correct answer → AI makes 3 distractors → later solve as a 4-option MCQ.

Rapid Revision: Open 'Due Today' → mixed image + MCQ session based on previous performance.

16. DEVELOPMENT ACCEPTANCE CRITERIA

- User can create an image card and retrieve it later.
- Answer is not visible before the reveal action.
- User can create an MCQ using one supplied correct answer.
- System produces exactly three distractors for the standard 4-option MCQ.
- Correct answer is not exposed before submission.
- User can edit AI-generated distractors.
- Cards persist after closing/reopening the app.
- Saved cards can be revised without re-entering their original content.
- Search/filter and revision sessions work.
- Review history is saved and can influence future scheduling.

17. SUGGESTED APP SCREENS

- Home
- Create Card
- Image Card Creation
- MCQ Card Creation
- My Cards / Library
- Decks
- Revision
- Card Review Screen

- Statistics
- Settings / Backup / Account

18. MOST IMPORTANT REQUIREMENT

The app must treat creation and revision as separate processes. A flashcard created once becomes a reusable learning object. The user can open it later, recall the answer without seeing it, reveal/submit, record the result, and repeat the process as many times as needed.

19. FUTURE-READY ARCHITECTURE

- The flashcard database should support text, images, MCQs, explanations, tags, review history and scheduling from the beginning.
- Future AI features can include automatic question extraction from notes/PDFs, image-assistance, automatic tagging, explanation generation and weak-topic detection.
- Architecture should allow AI services to be upgraded independently from the core card database.

Document purpose: This PRD can be shared directly with the UI/UX designer, mobile developer, backend developer and AI integration developer as the functional baseline for Version 1.